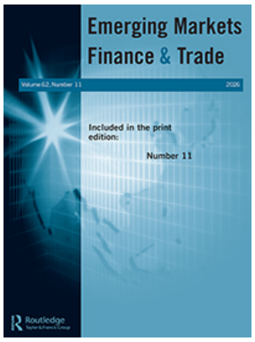

Routledge
Taylor & Francis Group

Emerging Markets Finance and Trade

ISSN: 1540-496X (Print) 1558-0938 (Online) Journal homepage: www.tandfonline.com/journals/mree20

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EMERGING MARKETS FINANCE AND TRADE
https://doi.org/10.1080/1540496X.2026.2691898

Navigating the AI–TFP Nexus: Internal Control Quality as a Key Mediator in Chinese Listed Firms

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ABSTRACT

While artificial intelligence (AI) is widely recognized for its transformative potential, the precise pathways through which it enhances firm productivity need further exploration. This research investigates the impact of AI adoption on total factor productivity (TFP) within Chinese A-share listed companies from 2007 to 2022. We find a significant positive effect of AI on firm-level TFP, a result that holds under a series of rigorous robustness checks. Importantly, we identify a key transmission mechanism: the positive relationship between AI and TFP is partially channeled through enhanced internal control quality (ICQ). This indicates that AI-driven improvements in organizational governance and risk management constitute a vital link to TFP gains. These findings highlight that the strategic integration of AI with robust ICQ systems is crucial for firms realizing their full productivity-enhancing potential, especially those in emerging markets.

KEYWORDS

Artificial intelligence; total factor productivity; Internal Control Quality

JEL

D24; G32; M42; O33; G34; C23

Introduction

Artificial intelligence (AI) has rapidly become a central force reshaping firm behavior, competitive dynamics, and economic performance. AI fundamentally alters how firms organize production and deploy resources (Mihet and Thomas 2019). However, despite the consensus on AI's transformative potential, a critical research question remains inadequately explored: through what specific mechanisms does AI adoption translate into measurable firm-level productivity gains? Recent studies emphasize that firms capable of effectively integrating AI technologies tend to gain significant competitive advantages, particularly in innovation efficiency and operational performance (Babina et al. 2024; Czarnitzki, Fernández, and Rammer 2023; Qin et al. 2024). AI is framed as an innovation that boosts firm total factor productivity (TFP), the standard gauge of long-term productivity growth and economic efficiency (Dwivedi et al. 2021). However, a critical research question remains inadequately explored: through what specific mechanisms does AI adoption translate into measurable firm-level productivity gains? This gap motivates our study to move beyond establishing a direct link and to uncover the underlying channels, particularly internal governance pathways, that facilitate this transformation.

A stream of studies argues that the impact of AI on TFP may be limited or delayed. AI technologies have not yet reached sufficient maturity, may generate organizational friction, or could lead to labor displacement that offsets efficiency gains, thereby weakening their impact on TFP (Gordon 2018). Meanwhile, previous studies suggest that AI enhances production efficiency by improving process control, product quality, and resource allocation, as well as by complementing automation and human capital (Ballestar et al. 2021; Damioli, Van Roy, and Vertesy 2021; Duan et al. 2023; Trajtenberg 2019). While these studies provide important insights, much of the debate remains focused on whether AI raises productivity, rather than how AI translates to measurable firm-level TFP gains.

This study argues that internal control quality (ICQ) constitutes a critical but underexplored mechanism linking AI adoption to TFP. High-quality internal control systems help align technological investments with corporate strategy, mitigate operational risks, and ensure effective implementation of advanced

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Supplemental data for this article can be accessed online at https://doi.org/10.1080/1540496X.2026.2691898.

technologies; thus ICQ can mediate the translation of AI investments into productivity gains. Prior research shows that high-quality internal control systems are essential for aligning technological investments with corporate strategy, mitigating operational risks, and ensuring effective implementation of advanced technologies (Liu, Xu, and Tao 2024). Moreover, governance structures embedded in internal control environments influence both the adoption and utilization efficiency of AI, thereby shaping its ultimate impact on firm performance (Zhang and Dong 2023). Building on this insight, we propose and test the hypothesis that ICQ is a key mediating mechanism in the AI–TFP relationship. Empirically identifying this mediating role allows us to move beyond the question of whether AI affects TFP.

To test this, we empirically examine the impact of AI on TFP and the mediating role of ICQ using a sample of Chinese A-share listed firms from 2007 to 2022. Mediation results attribute part of the productivity improvement to enhanced internal controls. The positive impact is particularly evident among non-state-owned, larger, non-manufacturing, and technology-intensive firms. China provides an ideal empirical setting for this analysis. Its rapid digital transformation, sustained investment in AI technologies, and distinctive corporate governance framework create a unique environment in which the interaction between AI adoption and internal governance can be rigorously examined. Insights drawn from this context are therefore informative not only for China but also for other emerging and developing economies undergoing similar digital transitions.

The motivation for this hypothesis is twofold. First, from a theoretical perspective, high-quality internal control systems are essential for aligning technological investments with strategy, mitigating operational risks, and ensuring the effective implementation of advanced technologies (Liu, Xu, and Tao 2024). We argue that AI adoption likely enhances information processing, monitoring, and reporting capabilities, which are core components of ICQ. In turn, improved ICQ can strengthen resource allocation, reduce inefficiencies, and solidify the foundation for AI-driven process improvements, thereby channeling AI's potential into realized productivity gains. Second, from an empirical standpoint, the role of governance structures, specifically ICQ, in conditioning the payoff from digital and AI investments is highlighted in emerging literature (L. Chen, Yang, and Yu 2025; Zhang and Dong 2023), yet a systematic investigation of ICQ as a mediator between AI and TFP is missing.

Our approach involves constructing a firm-level measure of AI adoption using machine-learning-based textual analysis of annual reports and employing the Olley–Pakes method to estimate TFP. We utilize a high-dimensional fixed-effects model for baseline estimation and a formal mediation analysis framework to test the indirect effect. Our findings confirm that AI adoption has a significant positive effect on firm TFP. More importantly, we find robust evidence that this effect is partially mediated by enhancements in internal control quality, confirming our central hypothesis. Heterogeneity analysis further reveals that the positive effect of AI on TFP is more pronounced in non-state-owned, larger, and technology-intensive firms.

This study makes three primary contributions to the literature, each addressing a distinct gap.

First, it shifts the empirical focus from the macro to the micro level and from correlation to mechanism. A substantial body of research has documented a positive association between AI adoption and productivity growth, yet this literature remains predominantly macro-oriented. Studies by Trajtenberg (2019), Damioli, Van Roy, and Vertesy (2021), and Czarnitzki, Fernández, and Rammer (2023) operate at the industry or country level, leaving the firm-level "black box" largely unopened. At the micro level, existing firm-level work (e.g., Babina et al. 2024; Wu and Zhu 2025; Yang 2022) has largely established direct correlations between AI investment and output metrics, without illuminating the organizational mechanisms through which AI is converted into productive efficiency. We address this gap by shifting the empirical focus from *correlation* to *mechanism*. Specifically, we identify and empirically validate ICQ as a significant transmission channel through which AI adoption translates into firm-level TFP gains. By doing so, we answer the "how" question that prior literature has left underexplored: AI does not automatically enhance productivity; its impact is mediated by the quality of internal governance structures.

Second, we integrate AI, ICQ, and TFP into a unified analytical framework. The literature on AI–TFP and the literature on digitalization and internal governance have developed largely in parallel. The former focuses on technology-driven efficiency gains (Culot, Podrecca, and Nassimbeni 2024; Haefner et al. 2021; Kromann et al. 2020), while the latter examines governance quality as either an outcome of digital transformation or a standalone determinant of firm performance (Tang and Tang 2025; K. Wang et al.

2023; Zhang and Dong 2023). These streams rarely intersect. Our study bridges this divide by theorizing and demonstrating that internal governance is not merely an outcome or a separate factor, but a *conduit* for technological impact. We integrate AI, ICQ, and TFP into a unified analytical framework, offering a more complete picture of the organizational prerequisites for successful AI integration. Our findings highlight that the productivity returns to AI are contingent on the quality of internal governance structures, which is a contingency that neither literature stream has adequately theorized.

Third, from a methodological perspective, we construct a novel, firm-specific measure of AI exposure. Existing micro-level research on AI adoption has relied heavily on proxies such as patent counts (K. Li et al. 2021), robotics data (Duan et al. 2023), or industry-level AI intensity measures. While useful, these proxies suffer from well-known limitations: patents capture invention rather than adoption; robotics data miss software-based AI applications; and industry-level measures cannot distinguish between adopters and non-adopters within the same sector. We address these measurement constraints by constructing a novel, firm-specific measure of AI exposure using text analysis of corporate annual reports, following W. Chen and Srinivasan (2024). This approach captures a broader spectrum of AI adoption activities, including software, algorithms, and cognitive applications that traditional proxies overlook. The resulting measure offers a more precise and comprehensive proxy for AI engagement at the firm level and provides a replicable tool for future micro-level research in contexts where traditional measures are limited.

Literature Review and Hypotheses

AI–TFP Nexus

Insights from financial and governance literature clarify the channels through which AI may boost TFP. Under agency theory (Jensen and Meckling 1976), asymmetric information between executives and shareholders can lead to poor investment and operational outcomes that reduce efficiency. AI, by improving data collection, processing, and decision-making, reduces information gaps and monitoring costs, allowing firms to allocate resources more efficiently and enhancing TFP (L. Chen, Yang, and Yu 2025; Wu and Zhu 2025). In line with resource-based theory (Barney 1991), AI can be considered a strategic intangible asset that strengthens a firm's capabilities to combine capital, labor, and knowledge in productive ways.

Firms leveraging AI technologies can optimize production processes, reduce operational inefficiencies, and enhance output relative to inputs, directly raising TFP (Babina et al. 2024; W. Li et al. 2024). AI deepens knowledge capital and constitutes an important driver of firm-level TFP. Unlike conventional technologies, AI is characterized by strong scalability and wide applicability, enabling its integration across core corporate activities (Kromann et al. 2020). Beyond technological innovation, AI-enabled supply chain management reduces inventory inefficiencies, resource waste, and operational losses, improving operational efficiency and contributing to productivity gains (Culot, Podrecca, and Nassimbeni 2024). AI also strengthens data-driven decision-making by improving firms' ability to forecast market demand, production conditions, and macroeconomic trends (Rana et al. 2022). By mitigating information-processing constraints and facilitating more effective organizational coordination (Haefner et al. 2021; Yang 2022), AI supports productivity gains through innovation, efficiency improvement, and organizational upgrading, forming a direct channel from AI Thus, the hypothesis is proposed:

H1: AI positively affects TFP.

Role of ICQ in AI and TFP

ICQ functions as a mediating governance mechanism that translates technological or strategic inputs such as AI adoption into performance outcomes by improving monitoring, reporting, and operational reliability. ICQ also stabilizes firm performance under environmental uncertainty, guiding managerial decisions and safeguarding resources, which amplifies the productivity benefits of AI (Saba and Ngepah 2024; K. Wang et al. 2023; Zhang and Dong 2023). High-quality internal controls facilitate the effective transformation of technological inputs into productivity gains (X. Wang and Kong 2025). Consistent with the intelligent governance perspective, AI adoption enhances internal governance structures by improving information processing and control effectiveness, thereby supporting productivity improvements (L. Chen, Yang, and Yu 2025).

Corporate governance theory suggests that strong ICQ ensures AI-generated insights are properly implemented and monitored, reducing operational and compliance risks (Morris 2011; Tang and Tang 2025; X. Wang and Kong 2025). By enhancing organizational information-processing efficiency, AI functions to reduce data silos and facilitate the timely generation, transmission, and analysis of critical information, thereby strengthening ICQ and improving TFP (W. Li et al. 2024). High-quality ICQ ensures that AI-enhanced processes and managerial practices are effectively implemented, thereby strengthening the impact of AI on TFP positively. Therefore, the hypothesis is proposed:

H2: Internal control quality (ICQ) partially mediates the effect of AI on TFP.

Data and Methodology

Data Sources

The dataset includes Chinese A-share listed firms on the Shanghai and Shenzhen markets for 2007–2022. Firm characteristics were sourced from CSMAR, firms' annual/CSR reports, and websites. ICQ values are from the Dib Internal Control Database.

Variables

Dependent Variable: TFP

TFP captures overall firm productivity by jointly weighting inputs and outputs and is widely used to proxy technological progress and efficiency. *TFP* is obtained using the Olley and Pakes (1996) procedure, a semi-parametric technique that exploits investment decisions to control for simultaneity between inputs and productivity. Taking logarithms yields:

$$InY_{it} = \alpha InL_{it} + \beta InK_{it} + u_{it} \tag{1}$$

where Y_{it}, L_{it} and K_{it} denote output, labor, and capital, respectively, with the residual u_{it} interpreted as log-TFP.

Following Olley and Pakes (1996) and incorporating export decisions as in Löcker (2007), the production function is specified as:

$$InY_{it} = \beta_0 + \beta_k InK_{it} + \beta_l InL_{it} + \beta_a age_{it} + \beta_s state_{it} + \beta_e EX_{it} + \sum_m \delta_m year_m + \sum_m \lambda_n reg_n + \sum_\zeta \zeta_k ind_k + \varepsilon_{it} \tag{2}$$

To address endogeneity and sample selection bias, the three-step Olley—Pakes estimator refined by Melitz and Polanec (2015) is applied, accounting for capital endogeneity and firm exit.

Independent Variable: AI

Following W. Chen and Srinivasan (2024) and the WIPO AI glossary, *AI* is identified via textual analysis of Chinese AI terms measured as ln(1+ the count of AI-related words), so larger values indicate greater AI use.[1]

Mediator: ICQ

ICQ is measured using Dib Internal Control Index (Bo and Li 2025) which is 0–1000 normalized by dividing by 100 yielding a 0–10 scale, with higher values indicating better ICQ.

Control Variables

To capture firm characteristics that may affect corporate governance and TFP, firm size (*Size*), leverage (*Lev*), return on assets (*ROA*), firm age (*Age*), revenue growth (*Growth*), ownership concentration (*Share*), book-to-market ratio (*BM*), and the tangible assets ratio (*TAR*) are included as *Controls*.[2]

Methodology

Baseline Model

The baseline specification uses a high-dimensional fixed-effects framework to identify AI's impact on firm TFP specified as:

$$TFP_{it} = \Upsilon_0 + \Upsilon_1 AI_{it} + \sum_j \Upsilon_j controls_{it} + \varepsilon_{it} \tag{3}$$

$$TFP_{it} = \Upsilon_0 + \Upsilon_1 AI_{it} + \sum_j \Upsilon_j controls_{it} + \mu_t + \lambda_t + \eta_{s(t)} + \rho_{p(t)} + \varepsilon_{it} \tag{4}$$

where TFP_{it} denotes total factor productivity of firm i in year t, AI_{it} captures firm-level AI adoption, $Controls_{it}$ and ε_{it} represent control variables and the error term μ_i. λ_t, $\eta_{s(t)}$ and $\rho_{p(t)}$ are firm, year, industry, and province fixed effects.

Mediation Model

A mediation model explains how an independent variable affects an outcome through a mediator, capturing its indirect effect. The mediation can be full or partial, depending on whether the direct effect remains significant after including the mediator (Baron and Kenny 1986). To examine whether ICQ mediates the relationship between AI and TFP, the following models are estimated based on equation (4).

$$ICQ_{it} = \delta_0 + \delta_1 AI_{it} + \sum_j \delta_j Controls_{it} + \lambda_t + \eta_{s(t)} + \rho_{p(t)} + \varepsilon_{it} \tag{5}$$

$$TFP_{it} = \theta_0 + \theta_1 AI_{it} + \theta_2 ICQ_{it} + \sum_j \theta_j Controls_{it} + \mu_t + \lambda_t + \eta_{s(t)} + \rho_{p(t)} + \varepsilon_{it} \tag{6}$$

Equation (5) captures the effect of AI on ICQ, while Equation (6) estimates both the direct effect of AI on TFP and the indirect effect via ICQ. All models employ panel fixed effects at the firm, year, industry, and provincial levels to control for unobserved heterogeneity.

Empirical Analysis

Descriptive Statistics

Table 1 reports summary statistics for the main variables, with emphasis on TFP, AI, and ICQ.

Results of Baseline Regression

Across the baseline models reported in Table 2, *AI* carries a positive sign and is significant, with point estimates spanning 0.0300 to 0.0690 and p-values corresponding to the 1% or 5% levels. The Hausman test strongly rejects random effects, motivating the fixed-effects specification.[3] After sequentially including fixed effects, the impact of *AI* on *TFP* persists, though the magnitude decreases slightly in the most stringent specification in column 4. This robustness indicates that the productivity gains associated with *AI* are not driven by omitted variables or firm-specific characteristics but rather reflect a genuine causal relationship. These results support hypothesis H1, consistent with Yang (2022), Trajtenberg (2019), and Damioli, Van Roy, and Vertesy (2021), confirming that AI adoption enhances firm-level TFP.

Table 1. Descriptive statistics.

Variables	Observations	Mean	standard deviations	Minimum	Median	Maximum
TFP	38,300	7.0468	0.9321	3.4010	6.9397	11.6800
AI	38,300	0.7421	1.1513	0.0000	0.0000	4.8752
ICQ	38,300	6.4271	1.3109	0.0000	6.6674	9.6840
Size	38,300	22.1939	1.2896	19.4350	22.0076	26.5105
Lev	38,300	0.4366	0.2046	0.0389	0.4319	0.9385
ROA	38,300	0.0342	0.0693	−0.6203	0.0354	0.2355
Age	38,300	2.8201	0.3764	1.0986	2.8904	3.5835
Growth	38,300	0.1729	0.4368	−0.6735	0.1080	5.2226
Share	38,300	57.3987	15.2864	19.3207	58.0932	91.2196
BM	38,300	0.6161	0.2490	0.0559	0.6118	1.2451
TAR	38,300	0.9283	0.0853	0.4496	0.9568	1.0000

Table 2. Results of baseline regression.

	(1)	(2)	(3)	(4)
	TFP	*TFP*	*TFP*	*TFP*
AI	0.0300***	0.0690***	0.0345***	0.0147**
	(0.0098)	(0.0081)	(0.0065)	(0.0060)
Controls	No	No	Yes	Yes
Observations	38,300	37,820	38,300	37,820
R^2	0.0014	0.8292	0.5730	0.8870
Adjusted R^2	0.0013	0.8086	0.5729	0.8734
Firm FE	No	Yes	No	Yes
Year FE	No	Yes	No	Yes
Industry FE	No	Yes	No	Yes
Province FE	No	Yes	No	Yes

Note: Standard errors in parentheses; ***$p < .01$; **$p < .05$.

Robustness Checks

Alternative TFP

We utilize the Levinsohn–Petrin method (Levinsohn and Petrin 2003) to calculate an alternative measure of TFP (*TFP_LP*). When replacing *TFP* with *TFP_LP*, the coefficients of AI remain consistently positive and highly significant at the 1% level across all specifications in Table 3 columns (1) to (4). The magnitudes ranging from 0.0261 to 0.1007 are comparable to the baseline results, confirming the positive effect of adoption of AI on TFP.

Narrow Study Period

Restricting the sample to 2012–2019, thereby excluding periods affected by the 2008 global recession, the 2011 Eurozone debt turmoil, and the COVID-19 crisis, yields coefficients of 0.0613 and 0.0156 in columns (5) and (6). Each estimate is significant at 1%, which supports the conclusion that AI continues to boost TFP when crisis-affected years are excluded.

Table 3. Robustness checks.

	Alternative TFP				Narrow study period		Alternative AI	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	TFP_LP	*TFP_LP*	*TFP_LP*	*TFP_LP*	*TFP*	*TFP*	*TFP*	*TFP*
AI	0.0646***	0.1007***	0.0673***	0.0261***	0.0613***	0.0156***		
	(0.0117)	(0.0090)	(0.0065)	(0.0056)	(0.0081)	(0.0060)		
AI_alt							0.1077***	0.0223*
							(0.0164)	(0.0121)
Controls	No	No	Yes	Yes	No	Yes	No	Yes
Observations	38,300	37,820	38,300	37,820	20,090	20,090	37,702	37,702
R^2	0.0046	0.8583	0.6942	0.9255	0.8854	0.9221	0.8288	0.8872
Adjusted R^2	0.0046	0.8412	0.6941	0.9165	0.8623	0.9063	0.8081	0.8735
Firm FE	No	Yes	No	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	No	Yes	Yes	Yes	Yes	Yes
Industry FE	No	Yes	No	Yes	Yes	Yes	Yes	Yes
Province FE	No	Yes	No	Yes	Yes	Yes	Yes	Yes

Note: Standard errors in parentheses; ***$p < .01$; *$p < .10$.

Alternative AI

To address document length bias, we add a length standardized measure (*AI_alt*). AI_alt is denotes the natural log of one plus the AI keyword count standardized per 10,000 words. $AI\_alt = \ln(1 + \frac{\text{AI keyword count}}{\text{total words}} \times 10000)$. *AI_alt* as a proxy for AI adoption, the results show positive and significant coefficients of 0.1077 and 0.0223 in columns (7) and (8). This alternative indicator captures capital-intensive automation and again demonstrates a robust positive association with TFP. Moreover, the use of a length-standardized version of the narrow dictionary (*AI_narrow*) yields qualitatively similar results.[4]

The robustness checks indicate that the productivity gains tied to AI are robust to alternative ways of measuring TFP, alternative AI proxies, and varying sample periods. These findings provide strong empirical support for **H1**, underscoring that AI adoption significantly enhances firm-level productivity.

Endogeneity Test

To address potential endogeneity between AI adoption and TFP, a two-stage least squares following Roberts and Whited (2013) is applied. As an instrument, we employ the industry-year average AI intensity excluding the focal firm ($\overline{AI}$). This instrument $\overline{AI}$ is likely correlated with a firm's AI adoption because firms within a given industry are subject to comparable technological landscapes, competitive dynamics, and regulatory frameworks, which shape AI uptake. At the same time, the industry mean $\overline{AI}$ constructed without the firm's own observation which operates at the industry level and therefore should not directly determine an individual firm's TFP, satisfying the exogeneity requirement. The leave-one-out construction also prevents mechanical correlation and enhances the credibility of the instrument.

Table 4 shows the results of endogeneity test. The first-stage estimates indicate a strong and statistically significant relationship between the instrument and firm-level AI at 1% level. In the second-stage IV regression, the AI coefficient is estimated at 0.0803 and is significant at the 5% level. Diagnostic tests further validate the instrument, indicating that the instrument is not weak. These diagnostics verify that the IV strategy successfully removes endogeneity and that the estimates are reliable.

Meanwhile, $\overline{AI}$ may be presumed to satisfy the exclusion restriction, as it is unlikely to correlate with idiosyncratic, firm-level TFP shocks. To further isolate the effect of interest, the second-stage estimation incorporates a full set of industry × year fixed effects to control for unobserved heterogeneity across industries and time. The estimation subsumes the instrument, which suggests that, while industry shocks may exist, they do not completely invalidate the IV strategy or overturn the core finding.[5]

Mediation Analysis

Table 5 reports the mediation analysis assessing the role of ICQ in linking AI to TFP. The findings indicate that AI positively relates to ICQ, with an estimated coefficient of 0.0265 significant at the 5% level. When

Table 4. $\overline{AI}$ as IV.

	The first stage	The second stage
	AI	*TFP*
$\overline{AI}$	0.8070***	
	(0.0522)	
AI		0.0803**
		(0.0391)
Controls	Yes	Yes
Observations		37,809
R^2		0.3372
Adjusted R^2		0.3359
Firm FE		Yes
Year FE		Yes
Industry FE		Yes
Province FE		Yes
Kleibergen-Paap rk LM statistic		75.063***
Cragg-Donald Wald F statistic		1506.648

Note: Standard errors in parentheses; ***$p < .01$; **$p < .05$.

Table 5. Results of ICQ as a mediator.

	(1)	(2)
	ICQ	*TFP*
AI	0.0265**	0.0144**
	(0.0126)	(0.0060)
ICQ		0.0133***
		(0.0060)
Controls	Yes	Yes
Observations	38,300	38,300
R^2	0.4159	0.8872
Adjusted R^2	0.3454	0.8736
Firm FE	Yes	Yes
Year FE	Yes	Yes
Industry FE	Yes	Yes
Province FE	Yes	Yes
Sobel	0.0004*	
	(0.0002)	
Goodman-1 (Aroian)	0.0004*	
	(0.0002)	
Goodman-2	0.0004*	
	(0.0002)	
Bootstrap	0.0004**	
	(0.0138)	
confidence interval	[0.0000364, 0.0006718]	

Note: Standard errors in parentheses; *** $p < .01$; ** $p < .05$; * $p < .10$.

ICQ is incorporated into the TFP regression together with AI, both variables retain positive and statistically significant coefficients for *AI* at 0.0144 and for *ICQ* at 0.0133, indicating that higher ICQ partially mediates the effect of AI on TFP. Sobel-Goodman tests confirm the mediation, with an indirect effect of 0.0004, contributing approximately 2.4% of the total effect, while bootstrap analysis shows a mediating effect of 0.0004 with a p-value of 0.029 and a 95% confidence interval excluding zero. These findings strongly support hypothesis H2, demonstrating that ICQ is a significant mechanism through which AI enhances productivity.

The results highlight that the ICQ is a critical factor for realizing the productivity gains from AI adoption. By improving ICQ, firms can not only leverage AI to streamline processes and reduce operational risks but also enhance resource allocation, operational efficiency, and overall agility. This underscores the strategic importance of strengthening internal control systems alongside AI implementation, as such improvements amplify the direct impacts of AI on TFP. Overall, the evidence confirms that ICQ plays a meaningful mediating role in translating AI investments into tangible productivity improvements, reinforcing the validity of H2.

Heterogeneity Analysis

Ownership Structure: State-Owned (SOE) Vs. Non-State-Owned (Non-SOE) Firms

To assess whether ownership structure moderates the relationship between AI and TFP, firms are grouped into *SOE* and *Non-SOE* categories (Jara, Musacchio, and Wagner 2024). The coefficient for AI in column (1) of Table 6 for *SOE* firms is 0.0007 and not statistically significant, indicating a limited impact of AI on TFP in these firms. In contrast, in column (2) of Table 6 *AI* positively and statistically significant affects TFP among *Non-SOE* firms, with a coefficient of 0.0206 in, providing support for H1 that AI improves productivity. This contrast suggests that *Non-SOE* firms are positioned to integrate AI effectively to gain TFP better, whereas *SOE* firms may need to strengthen AI implementation to achieve similar benefits.

The heterogeneity analysis highlights that AI positively affects TFP primarily in non-state-owned firms, reinforcing H1. Firm ownership shapes the effectiveness of AI investments, and targeted efforts to integrate AI may yield stronger productivity improvements in firms with more flexible governance structures.

Firm Scale: Large Vs. Small

Firms are classified as large or small according to the median level of total assets, enabling a more refined comparison across organizational scales, and the estimates in columns (3) and (4) of Table 6 indicate *AI*

Table 6. Results of heterogeneity analysis.

	SOE	*Non-SOE*	Large-scale	Small-scale	Manufacturing	Non-manufacturing	High-tech	Non-high tech
	(1) *TFP*	(2) *TFP*	(3) *TFP*	(4) *TFP*	(5) *TFP*	(6) *TFP*	(7) *TFP*	(8) *TFP*
AI	0.0007	0.0206***	0.0192**	0.0085	0.0133**	0.0240**	0.0220**	0.0123*
	(0.0111)	(0.0070)	(0.0082)	(0.0070)	(0.0062)	(0.0111)	(0.0098)	(0.0072)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	14,661	22,370	18,854	18,538	24,682	13,094	9809	27,978
R^2	0.8994	0.8832	0.8818	0.8298	0.9037	0.8888	0.8900	0.8927
Adjusted R^2	0.8887	0.8651	0.8656	0.7979	0.8907	0.8743	0.8734	0.8795
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: Standard errors in parentheses; ***$p < .01$; **$p < .05$; *$p < .10$.

positively and statistically significant increase in TFP among large firms, while the corresponding effect for small firms is weaker and statistically insignificant. Results suggest that large-scale firms can leverage AI more effectively to boost productivity, likely due to greater resources, advanced management practices, and operational capacity. In contrast, small-scale firms may face constraints in AI adoption or require more strategic integration to realize similar productivity gains. These findings underscore that firm scale is a critical factor in understanding how AI contributes to productivity, and that policy and managerial strategies should consider size-specific approaches to AI implementation.

Industry Type: Manufacturing Vs. Non-Manufacturing

Columns (5) and (6) of Table 6 distinguish between manufacturing and non-manufacturing firms to assess whether the effect of *AI* on *TFP* varies by industry. This classification facilitates a more granular evaluation of sector-specific drivers of *AI* and *TFP*. In column (5), the estimated coefficient on AI is 0.0133 and statistically significant at the 5% level, indicating a positive association for non-manufacturing firms. In column (6), the coefficient increases to 0.0240 and is significant at the 1% level, suggesting a comparatively stronger effect within manufacturing firms. The difference suggests that AI contributes more directly to productivity in manufacturing through process improvements and operational efficiencies. These results underscore the need for tailored AI integration strategies to maximize productivity gains across sectors.

Technological Intensity: High-Tech Vs. Non-High-Tech

Columns (7) and (8) of Table 6 report the heterogeneity analysis of AI's effect on TFP across high-tech and non-high-tech firms. For high-tech firms, the estimated coefficient on AI is 0.0220 and statistically significant at the 5% level, indicating a pronounced positive association with TFP. By contrast, the coefficient for non-high-tech firms is 0.0123 and only marginally significant at the 10% level, suggesting a weaker effect. These results suggest that AI contributes more substantially to productivity in high-tech sectors, likely due to the deeper integration of AI and advanced processes to maximize TFP gains.

Conclusion

This study demonstrates that AI significantly enhances TFP, highlighting its transformative role in improving operational efficiency. AI enhances TFP both directly and indirectly, with improvements in ICQ acting as a key channel through which technological adoption is converted into measurable efficiency gains. This underscores the importance of organizational context and governance structures in shaping the returns to AI adoption. By integrating AI with internal control systems, firms can improve decision-making, enhance risk management, optimize resource allocation, and strengthen operational resilience.

This study offers a frontier perspective for emerging economies, where effective governance mechanisms are crucial to harness technological innovations. By emphasizing the interaction among technological adoption, governance structures, and productivity outcomes, this study offers implications for managers seeking to advance high-quality corporate development while improving efficiency and transparency in emerging financial markets.

Notes

1. The AI seed words in annual report texts are treated as a corpus and expanded using the Word2vec Skip-gram model cosine similarity is used to select the ten closest terms for each seed word, and irrelevant, duplicate, and low-frequency terms are removed.
2. Table A.1 shows the definition of variables.
3. Results of Hausman test is in Appendix Table A.2.
4. The results are shown in Table A.3. *AI_narrow* is the textual AI measure constructed after removing infrastructure oriented terms (e.g., cloud, big data). Coefficients remain positive and generally significant, supporting robustness.
5. The results are shown in Table A.4.

Author Contributions

CRediT: **Mengling Lin:** Conceptualization, Data curation; **Chiu-Lan Chang:** Conceptualization, Supervision, Writing – original draft, Writing – review & editing; **Zhiyan Zheng:** Investigation, Methodology; **Ming Fang:** Funding acquisition, Project administration, Supervision, Validation.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Funding

This study was supported by Social Science Fund of Fujian Province of China [Grant number: FJ2024B019; FJ2025B043], Natural Science Foundation of Fujian Province of China [Grant number: 2024R0097], Fujian Provincial Department of Education China [Grant number: JAS24170; JAT241194], and Fuzhou Federation of Social Sciences [Grant number: 2024FZB37; 2025FZB175].

Availability of Data and Materials

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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